



MH3511: Data Analysis with Computer

Group Project

Uncovering the Housing Market: Size, Environment and Sale Prices

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1. Introduction

With over 2.3 billion houses globally sheltering over 7.9 billion people^[1], housing markets around the world show significant diversity in pricing influenced by much more than just square footage. Prospective buyers evaluate multiple considerations, including structural attributes (e.g., living area, roof style) and environmental factors (e.g., lot size, alley access) before making a purchase decision.

Understanding the drivers of housing prices is important for a wide range of stakeholders, including homeowners, investors, developers, agents, urban planners, and policymakers. Analysing the patterns in property features and their sale prices reveals the key factors determining market value, guiding better decisions that result in a more efficient housing market.

Our project analyses a dataset of house sale prices in Iowa, USA, during the 2010s, which includes variables such as property quality, room sizes, and the year of construction. Based on this data, we aim to address the following questions related to housing sales:

1. To what extent does overall property quality relate to sale price?
2. What is the correlation between house size (basement area, above-ground area, floor size) and sale price?
3. Is there a relationship between the quality of essential features (HeatingQC) and non-essential features (GarageFinish)?
4. Does the type of alley access (gravel or paved) affect the sale price of houses?
5. Does the style of roof influence sale price?
6. Does the year the house was sold (YrSold) have any effect on its sale price?
7. What linear relationships exist between different continuous variables and the sale price?
8. What are the key dimensions explaining variance in the dataset?

In this study, we examine the Iowa housing dataset using R language. For each research objective, we apply the most appropriate statistical methods to identify key patterns and relationships in the data. Our findings are backed by solid statistical evidence and clear explanations, enabling us to draw reliable conclusions.

2. Data Description

The dataset used in this analysis, titled "House Prices", was obtained from the Kaggle community and originally compiled by Dean De Cock in 2011 for educational use in data science.^[2] The data is publicly available and widely used for regression modeling and machine learning applications. Our analysis specifically uses the "train.csv" file.

To prepare the dataset for analysis, the following preprocessing steps were conducted:

1. **Variable Selection:**

A subset of relevant variables was selected to focus the analysis on meaningful

features related to house pricing. We focused on variables that directly impact home valuations based on:

- Practical Relevance: Selected features that homebuyers prioritise (e.g. living area, garage size) and appraisers typically consider (e.g. quality ratings)
- Data Quality: Avoided variables with overlapping and redundant information (e.g. Exterior1st and Exterior2nd, or multiple basement-related features).

The chosen variables collectively represent:

- Structural characteristics (e.g. GrLivArea, TotalBsmtSF)
- Quality assessments (e.g. OverallQual, HeatingQC)
- Property amenities (e.g. Fireplaces, GarageCars)
- Location factors (e.g. Alley, LotFrontage)

This focused approach balances analytical depth with interpretability while avoiding multicollinearity from overlapping measures.

2. New Variable Creation:

- TotalFloorSF was calculated by summing the 1st and 2nd floor square footage.
- YearBuiltDecade was created by binning YearBuilt into decade intervals (e.g. "1990s", "2000s") for easier analysis of historical patterns.

The final dataset consists of 1,460 observations (houses) and the following variables:

Discrete Numerical Variables:

1. OverallQual: Overall material and finish quality
2. GarageCars: Number of cars the garage can hold
3. Fireplaces: Number of fireplaces
4. KitchenAbvGr: Number of kitchens above grade
5. YrSold: Year the house was sold

Continuous Numerical Variables:

1. SalePrice: Amount house was sold for
2. GrLivArea: Above ground living area (sqft)
3. TotalBsmtSF: Total basement area (sqft)
4. LotFrontage: Length of the street connected to the property
5. LotArea: Lot size in square feet
6. TotalFloorSF: Combined area of 1st and 2nd floor

Categorical Variables:

1. HeatingQC: Heating quality and condition
2. RoofStyle: Type of roof
3. GarageFinish: Interior finish of the garage
4. BsmtFinType1: Rating of basement finished area
5. YearBuiltDecade: Decade the house was built
6. Alley: Type of alley access for the house

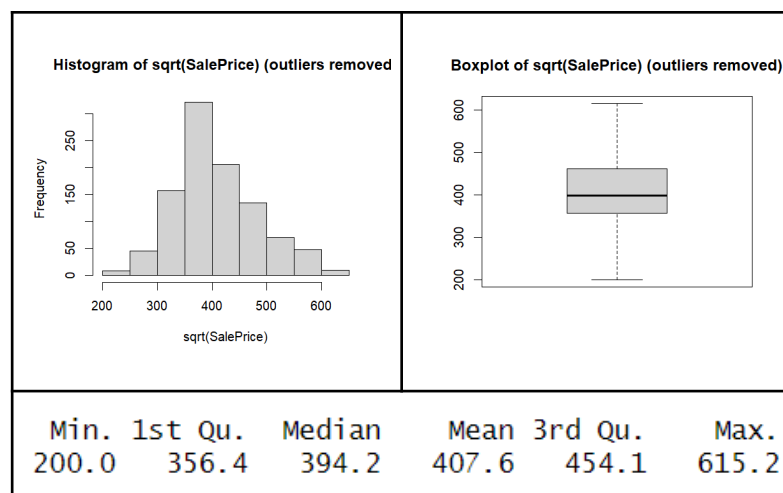
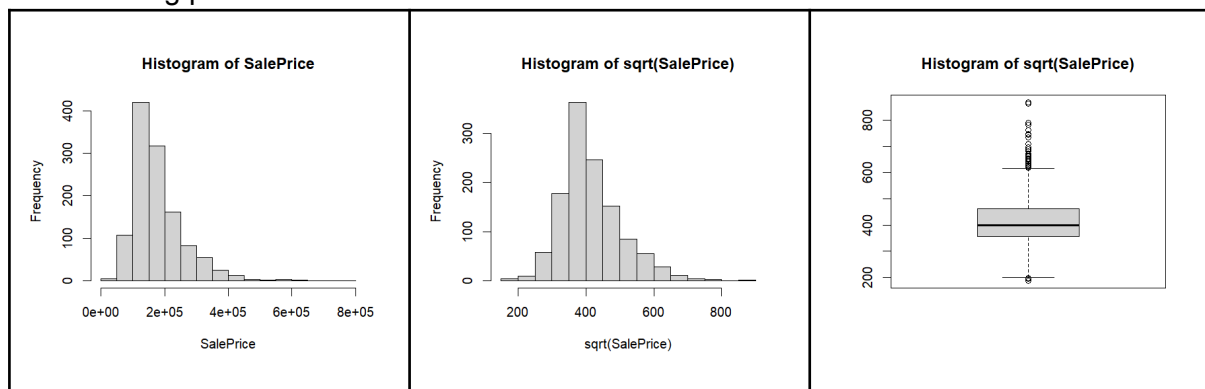
3. Description and Cleaning of Dataset

3.1 Removing incomplete data from dataset

After extracting the relevant variables, we investigated each variable to remove data with incomplete information and outliers. Firstly, for categorical variables *BsmtFinType1*, *GarageFinish* and *Alley*, “NA” data are replaced with “NoBasement”, “NoGarage” and “NoAlley” respectively. Subsequently, 259 data with “NA” for *LotFrontage* were removed from the dataset, retaining 1201 observations.

3.2 Summary statistics for the main variable of interest, *SalePrice*

The following plots show the overall distribution of the variable *SalePrice*.

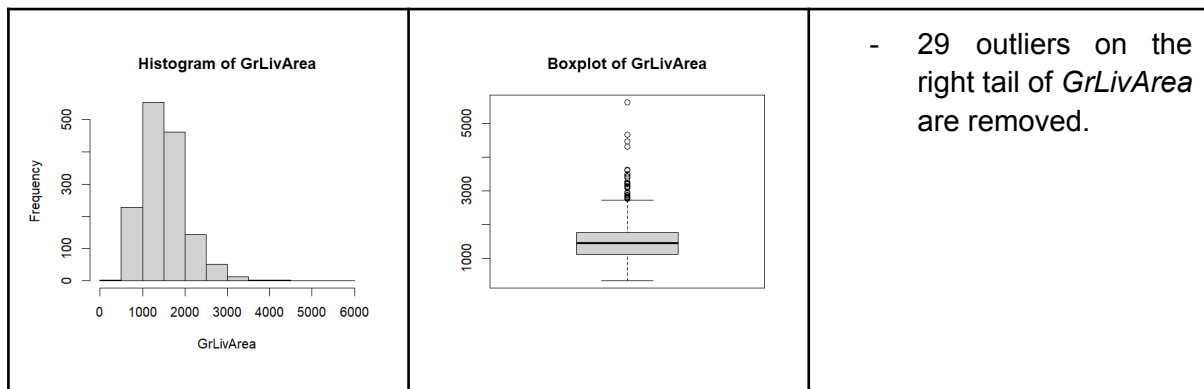


Given that the data in the variable *SalePrice* looks skewed, we applied a sqrt-transformation to *SalePrice*. Next, we utilised the interquartile range (IQR) method of outlier detection, whereby data points not within the subset of $(Q1 - 1.5 * IQR, Q3 + 1.5 * IQR)$ will be removed. This removes outlying data points from both the left and right tails of *sqrt(SalePrice)*, to achieve a better distribution. Similarly, the IQR method will be used for outlier removal of continuous variables in Section 3.3.

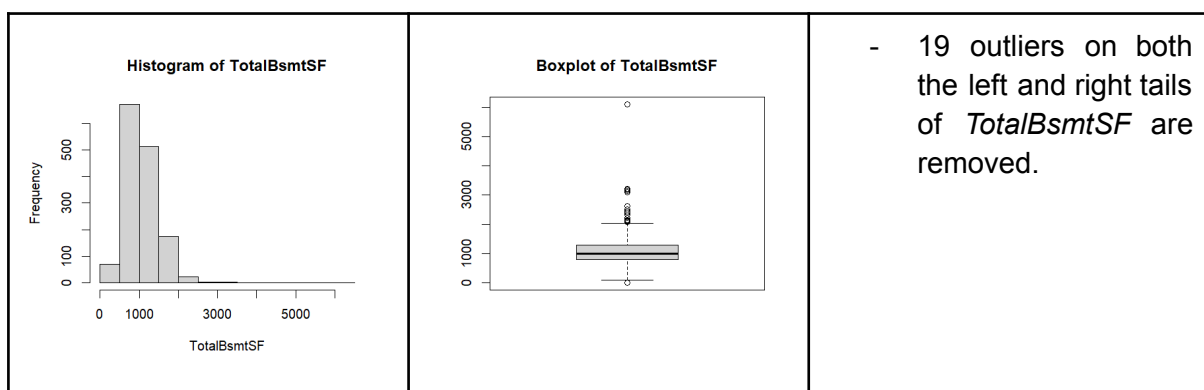
3.3 Summary statistics for continuous variables

The histogram, the boxplot, the transformation applied and the outliers removed from the variables are tabulated in the following subsections:

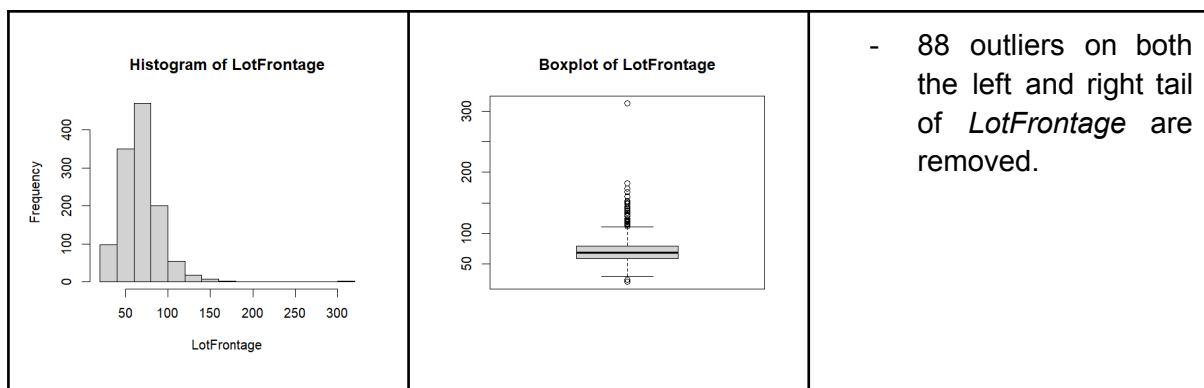
3.3.1 Above ground living area (sqft), *GrLivArea*



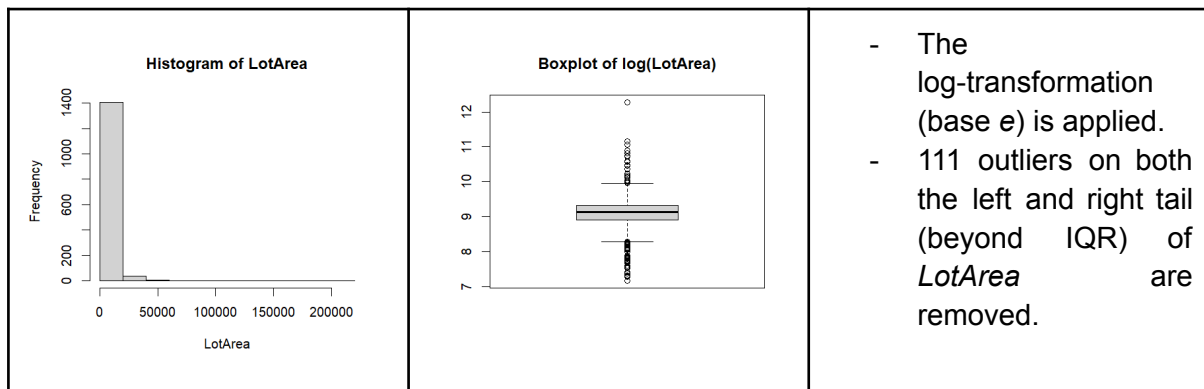
3.3.2 Total Basement Area (sqft), *TotalBsmtSF*



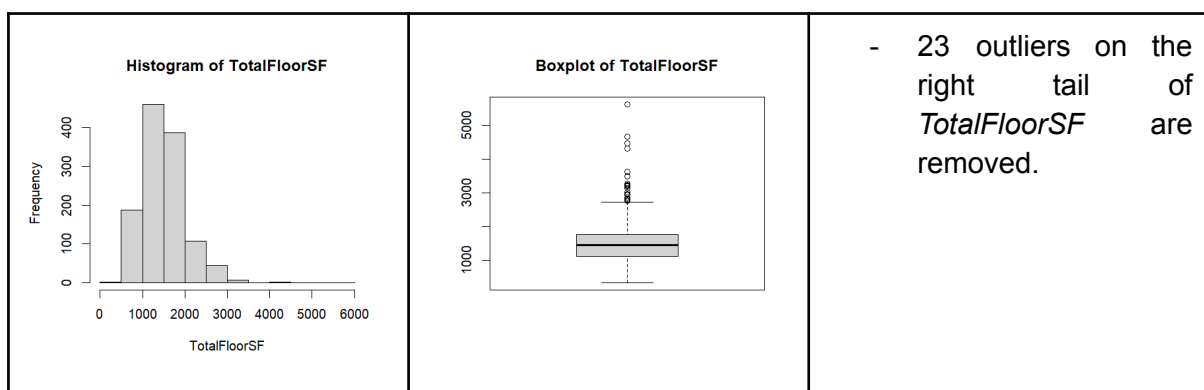
3.3.3 Length of the street connected to the property (sqft), *LotFrontage*



3.3.4 Lot size (sqft) with log-transformation, $\log(\text{LotArea})$

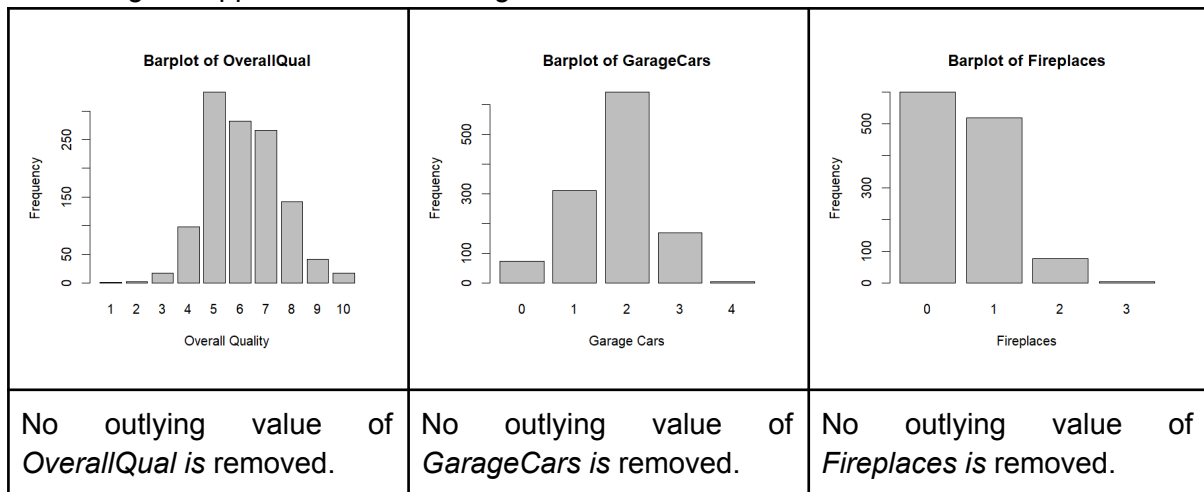


3.3.4 Combined Area of 1st and 2nd Floor (sqft), *TotalFloorSF*



3.4 Summary statistics for discrete variables

The histogram applied for the following discrete variables are tabulated as follows:



Barplot of KitchenAbvGr Frequency Number of Kitchen Above Grade	Barplot of YrSold Frequency Year Sold
No outlying value of <i>KitchenAbvGr</i> is removed.	No outlying value of <i>YrSold</i> is removed.

3.5 Summary statistics for the categorical variables

The histogram applied for the following categorical variables are tabulated as follows:

Barplot of HeatingQC Frequency HeatingQC	Barplot of RoofStyle Frequency RoofStyle	Barplot of GarageFinish Frequency GarageFinish
No outlying value of <i>HeatingQC</i> is removed.	No outlying value of <i>RoofStyle</i> is removed.	No outlying value of <i>GarageFinish</i> is removed.

Barplot of BsmtFinType1 Frequency BsmtFinType1	Barplot of YearBuiltDecade Frequency YearBuiltDecade	Barplot of Alley Frequency Alley
No outlying value of <i>BsmtFinType1</i> is removed.	No outlying value of <i>YearBuiltDecade</i> is removed.	No outlying value of <i>Alley</i> is removed. Only 86 houses in the dataset have an alley, and we will use this subset for further analysis.

3.6 Summary statistics for all variables

After accounting for all the outlying values within the above variables, here are the total outlying values removed from each variable. Given that some data may be an outlier for more than one variable, the summation of all the outliers in each variable may not necessarily add up to the total number of outlying data removed from the dataset.

<i>sqrt(SalePrice)</i>	<i>GrLivArea</i>	<i>TotalBsmtSF</i>	<i>LotFrontage</i>	<i>log(LotArea)</i>	<i>TotalFloorSF</i>
40	29	19	88	111	23


```

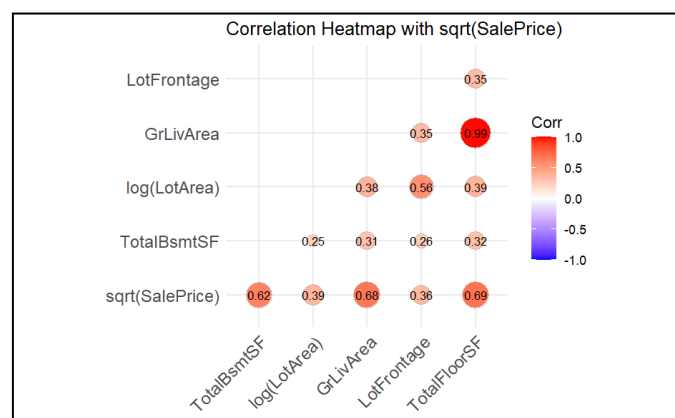
> colSums(is.na(df_outlier_filtered))
OverallQual      GarageCars      Fireplaces      LotFrontage      YearBuilt      YrSold      SalePrice
0              0              0              88              0              0              0
GrLivArea      TotalBsmtSF      KitchenAbvGr      HeatingQC      RoofStyle      GarageFinish      BsmtFinType1
29             19              0              0              0              0              0
TotalFloorSF    log(LotArea) sqrt(SalePrice) YearBuiltDecade
23             111             40              0

```

After removing the outlying values, the dataset retains a remaining of 1002 observations for further statistical analysis.

4. Statistical Analysis

4.1 Correlations between *sqrt(SalePrice)* and other Continuous Variables



A correlation heatmap is a color-coded visual representation of the correlation matrix. It helps us to analyse the possible linear relationships between *sqrt(SalePrice)* and other continuous specifications of houses.

With +1.0 indicating a perfect positive correlation and -1.0 representing a perfect negative correlation between two numeric variables, the correlation coefficients of 0.62, 0.68 and 0.69 suggests that a stronger positive linear correlation between *sqrt(SalePrice)* and *TotalBsmtSF*, *GrLivArea* and *TotalFloorSF* as compared to the other continuous variables.

Between the other continuous variables in this tabulation, we also observe some interesting relationships:

1. *GrLivArea* has a strong positive linear correlation with *TotalFloorSF* ($r=0.99$)

2. *TotalBsmntSF* has a weak positive linear correlation with *TotalFloorSF* ($r=0.32$)
3. $\log(\text{LotArea})$ has a moderate positive linear correlation with *LotFrontage* ($r=0.56$)

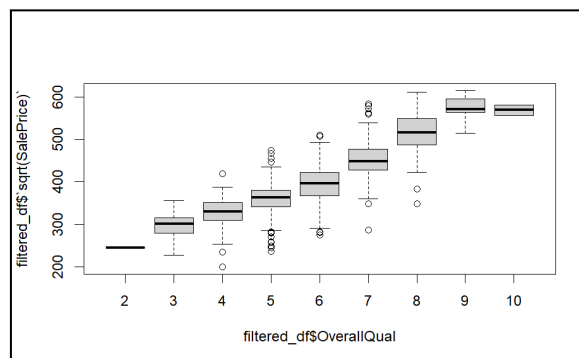
We will perform some statistical tests to further investigate some of these relationships in the next section.

4.2 Statistical Tests

In the previous section, we found a positive correlation between $\sqrt{\text{SalePrice}}$ and its size, including the size of the basement (*TotalBsmntSF*), above-ground living area (*GrLivArea*) and total floor size (*TotalFloorSF*). In this section, we seek to further investigate the relationships between $\sqrt{\text{SalePrice}}$ and qualitative and contextual aspects of a house and its sale (QAH), as well as between the QAH themselves.

4.2.1 Relation between $\sqrt{\text{SalePrice}}$ and *OverallQual*

In this section, we try to answer the question: “Does the overall quality of a house affect its sale price?”



A quick look at the boxplot reveals an increase of the median $\sqrt{\text{SalePrice}}$ as the grade of *OverallQual* increases. This confirms our further investigation into the specific features of a house that affect its quality.

4.2.2 Relation between *HeatingQC* and *GarageFinish*

Within a house, there exists essential and non-essential features. For example, *HeatingQC* which affects a family's quality of life in winter may be considered more essential compared to *GarageFinish*, which may be in less use.

In this section, we try to answer the question: “Is the quality of essential home features (*HeatingQC*) associated with the quality of non-essential features (*GarageFinish*)?”

To help us answer this question, we constructed a two-way contingency table between the two variables, then conducted a chi-squared test of independence between the two.

We test,

H_0 : There is no association between *HeatingQC* and *GarageFinish*

H_1 : There is an association *HeatingQC* and *GarageFinish*

	Ex	Fa	Gd	Po	TA
Fin	159	3	27	0	20
NoGarage	17	5	14	0	24
RFn	206	3	32	0	58
Unf	127	21	95	1	190

Upon generating the two-way contingency table, we realised that there was only one house in the dataset with poor heating quality, resulting in a column of mostly zeroes. To ensure the validity of the chi-squared test, we removed the 'Po' column before conducting the test.

	Ex	Fa	Gd	TA	
Fin	159	3	27	20	<pre>> chisq.test(BsmtFinType1vsHeatingQC_filtered) Pearson's Chi-squared test data: BsmtFinType1vsHeatingQC_filtered X-squared = 194.59, df = 9, p-value < 2.2e-16</pre>
NoGarage	17	5	14	24	
RFn	206	3	32	58	
Unf	127	21	95	190	

The chi-squared test returns a p-value of 2.2×10^{-16} . This demonstrates that the observed values of *HeatingQC* and *GarageFinish* differ significantly from their expected values under the assumption of independence. Therefore, at a significance level of 0.05, we reject H_0 and conclude that there exists an association between *HeatingQC* and *GarageFinish*.

4.2.3 Relation between *sqrt(SalePrice)* and *Alley*

In this section, we aim to find out the relationship between alley access types and the sale price of a house. The 86 houses with alleys in the dataset are classified into 2 alley access types, gravel and paved. We first conduct an F-test to check if the variances in the 2 groups are equal, followed by t-test to check whether the means of the 2 groups are equal.

<pre>> var.test(sqrt(SalePrice) ~ Alley, data = df_alley, conf.level = 0.9) F test to compare two variances data: sqrt(SalePrice) by Alley F = 0.39903, num df = 40, denom df = 22, p-value = 0.01134 alternative hypothesis: true ratio of variances is not equal to 1 90 percent confidence interval: 0.2058950 0.7238788 sample estimates: ratio of variances 0.3990283</pre>	<pre>> t.test(sqrt(SalePrice) ~ Alley, data = df_alley, var.equal=FALSE) Welch Two Sample t-test data: sqrt(SalePrice) by Alley t = -4.0139, df = 32.068, p-value = 0.0003358 alternative hypothesis: true difference in means between group Grv1 and group Pave is not equal to 0 95 percent confidence interval: -92.24626 -30.14295 sample estimates: mean in group Grv1 mean in group Pave 343.5727 404.7673</pre>
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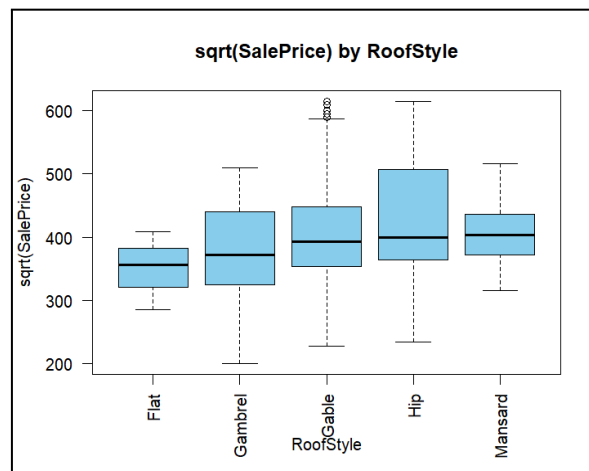
After conducting the F-test at 90% confidence interval which yields a p-value of 0.01134 (<significance level = 0.10), we reject H_0 (ratio of variances is equal to 1) and there is insufficient evidence to conclude that the variances of the 2 groups are equal. Hence, we used the non-equal variance two sample t-test at 95% confidence interval subsequently and since the p-value obtained is 0.0003358 which is significantly smaller than the significance level of 0.05, we reject H_0 (difference in means is equal to 0) and conclude that there is a statistically significant difference between the means of the group Grv1 and the group Pave.

In the output shown above, we can see that the mean *sqrt(SalePrice)* is higher for houses with paved alley access (404.7673) as compared to gravel (343.5727). This could possibly be attributed to paved alleys being perceived as having nicer aesthetics and easier for vehicles to navigate, or perhaps due to higher maintenance costs of gravel alleys which may deter buyers and result in a lower perceived value.

4.2.4 Relation between $\sqrt{\text{SalePrice}}$ and RoofStyle

Every house may have a roof, but not all roofs are made equal. Here, we investigate whether the style of a house's roof affects its sale price.

The roof types of the houses in our dataset fall into five categories, namely mansard, hip, gable, gambrel, and flat. An initial box plot of $\sqrt{\text{SalePrice}}$ by RoofStyle seems to suggest a relation between the two variables, with median $\sqrt{\text{SalePrice}}$ being the lowest for flat roofs and highest for mansard roofs:



We will now test this formally, using a one-way ANOVA test. We test:

$H_0: \mu_{\text{Flat}} = \mu_{\text{Gambrel}} = \mu_{\text{Gable}} = \mu_{\text{Hip}} = \mu_{\text{Mansard}}$ against $H_1: \text{not all } \mu_i \text{ are equal.}$

```
> summary(roofstyle_anova)
      Df Sum Sq Mean Sq F value    Pr(>F)
RoofStyle  4 121509    30377   5.463 0.000237 ***
Residuals 997 5543654    5560
---
```

In essence, since the p-value of 0.000237 is less than the significance level of 0.05, we reject H_0 and conclude that there are significant differences among the mean values of SalePrice for each roof type. In other words, there exists a relation between $\sqrt{\text{SalePrice}}$ and RoofStyle .

The houses used in our dataset originate from Ames, Iowa, a city which experiences a considerable amount of snowfall on a yearly basis as well as occasional severe thunderstorms with high-speed winds.^[3] For context,

- Hip roofs handle snow and wind the best among the roof types listed.^[4]
- Mansard roofs are a type of hip roof which is more aesthetically pleasing.^[5]
- Gable roofs handle snow well, but may be less suitable in high winds.^[6]
- Gambrel roofs provide additional space in the house, but are weaker in snowy or windy conditions.^[7]
- Flat roofs do not allow for snow drainage^[8] and should be avoided in Ames, Iowa, possibly accounting for the low popularity of homes with flat roofs (only three homes sold over five years).

Flat	Gable	Gambrel	Hip	Mansard
3	798	11	184	6

Some roof types are severely underrepresented, with flat, gambrel, and mansard roofs each having a small sample size under 30. Thus, we also used a non-parametric test, the Kruskal-Wallis Test, as an additional step since the earlier ANOVA analysis performed entailed assumptions of normality and equal variance among groups. We test:

H_0 : There is no significant difference in the distributions of the roof styles.

H_1 : There exists at least one roof style with higher sale prices than the rest.

```
Kruskal-Wallis rank sum test

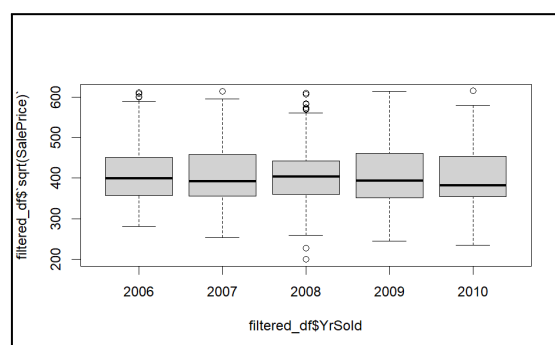
data:  sqrt(SalePrice) by RoofStyle
Kruskal-Wallis chi-squared = 11.99, df = 4, p-value = 0.01743
```

Similar to the one-way ANOVA, the Kruskal-Wallis test returned a p-value of 0.01743, which is less than the significance level of 0.05, and we reject H_0 , concluding that there is a significant difference in the mean $\sqrt{\text{SalePrice}}$ across roof styles. Despite the difference in what each test measures—ANOVA measures differences in mean, whereas Kruskal-Wallis test measures differences in distribution—both tests confirm our initial suspicion that the style of a house’s roof is indeed a factor related to its sale price.

4.2.5 Relation between $\sqrt{\text{SalePrice}}$ and YrSold

In the earlier section, we have observed how climate may affect the sale price of a house. In this section, we investigate further into the relationship between the temporal context (YrSold) and a house’s sale price.

We seek to answer the question, “Does the year when the house is sold affect its SalePrice?”



The boxplot of $\sqrt{\text{SalePrice}}$ by YrSold shows that the variances of $\sqrt{\text{SalePrice}}$ are similar across all years of sale. Thus, the one-way ANOVA test which operates under the assumption of equal variances is appropriate for testing if the means are equal.

We test,

$$H_0: \mu_{2006} = \mu_{2007} = \mu_{2008} = \mu_{2009} = \mu_{2010} \text{ against } H_1: \text{not all } \mu_i \text{ are equal}$$

```
> summary(aov(filtered_df$ sqrt(SalePrice) ~filtered_df$YrSold))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
filtered_df\$YrSold	4	9010	2253	0.397	0.811
Residuals	997	5656153	5673		

Since the p-value (0.811) returned by the ANOVA test is greater than 0.05, by the results of the ANOVA test, we do not reject H_0 at a 0.05 significance level and conclude that there is no significant difference in the means of $\sqrt{\text{SalePrice}}$ across the years sold.

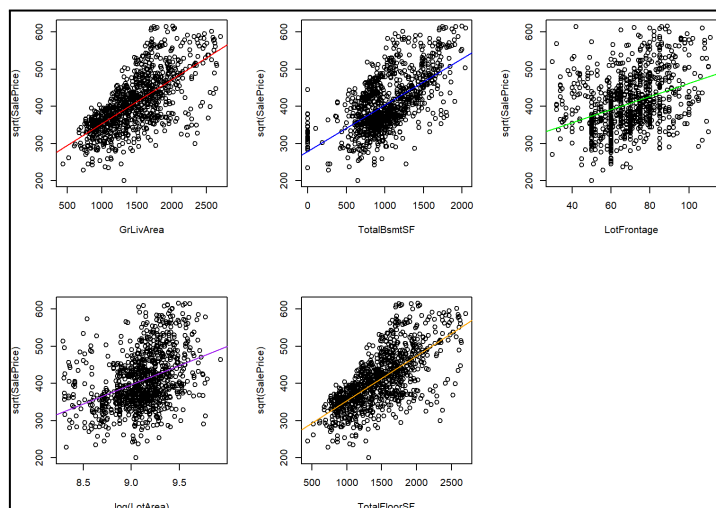
However, a closer look at the boxplot of $\sqrt{\text{SalePrice}}$ against YrSold reveals some interesting observations - while there are no significant differences in spread across the years of sale, $\sqrt{\text{SalePrice}}$ in 2008 seems to have a smaller variance than the other years. Additionally, the 2008 boxplot reveals two outliers below the minimum SalePrice which are not present in any other year, and a lower maximum SalePrice with the largest number of outliers at the right tail.

A further exploration leads us to find the most plausible reason for the abnormal data - the Global Financial Crisis. In addition to being a watershed event worldwide, this event was triggered by the Subprime Mortgage Crisis of the US^[9], where our data is sourced.

Hence, while there are no significant differences in the sale prices of different years looking at their mean alone, an analysis into their outliers, maximum and minimum values reveal that the events of different years still impact the sale prices.

4.2.6 Linear regression of continuous variables against $\sqrt{\text{SalePrice}}$

In section 4.1, we have already seen that the 5 continuous variables generally have a decent linear correlation with $\sqrt{\text{SalePrice}}$, and this can be supported by plotting the scatter plots with linear regression lines of the 5 continuous variables against $\sqrt{\text{SalePrice}}$.



We now perform a simple linear regression analysis to determine which of the 5 variables can be best used to model $\sqrt{\text{SalePrice}}$ linearly. The equation of linear regression model used is: $Y = \beta_0 + \beta_1 * X + \epsilon$, where $Y = \sqrt{\text{SalePrice}}$, X is one of the 5 variables, β_0 is the intercept parameter, β_1 is the slope parameter, and ϵ is the residual. Using the `summary()` function on the fitted linear regression model of $\sqrt{\text{SalePrice}}$ against GrLivArea , we can obtain the β_0 , β_1 , p-value and R^2 as shown below.

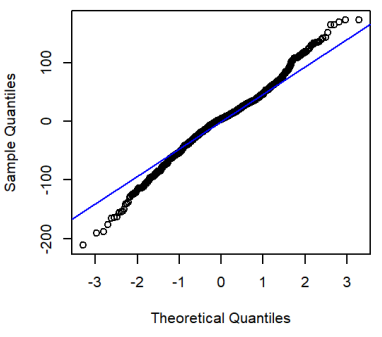
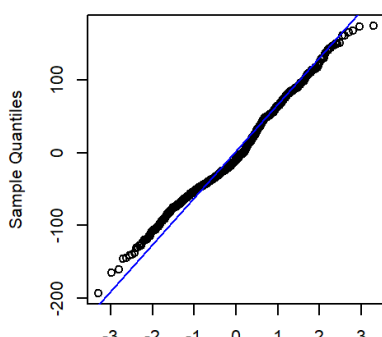
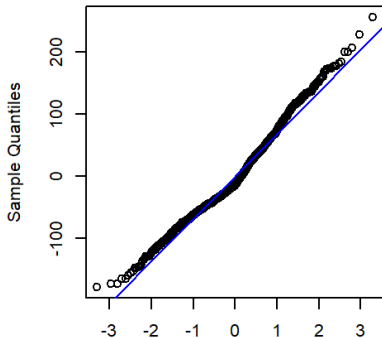
```
Call:
lm(formula = filtered_df$`sqrt(SalePrice)` ~ filtered_df$GrLivArea)

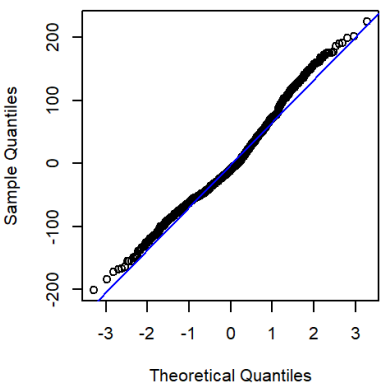
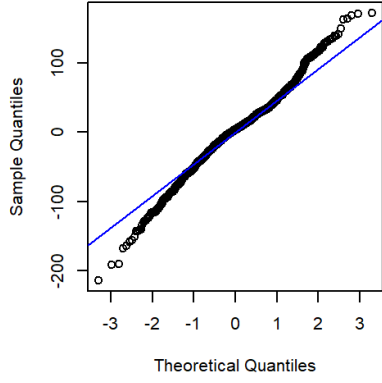
Residuals:
    Min       1Q   Median       3Q      Max
-211.787  -31.789   3.416   31.359  172.582

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.355e+02  6.151e+00  38.29  <2e-16 ***
filtered_df$GrLivArea 1.182e-01  4.051e-03  29.18  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 55.31 on 1000 degrees of freedom
Multiple R-squared:  0.46,    Adjusted R-squared:  0.4594
F-statistic: 851.8 on 1 and 1000 DF,  p-value: < 2.2e-16
```

We repeated the same steps for the other 4 variables and plotted the qq-plot of residuals to obtain the following table:

Variable (X)	Fitted Model	p-value	R^2	qq-plot of Residuals
<i>GrLivArea</i>	$\hat{Y} = 235.50 + 0.118X$	2.2×10^{-16}	0.460	
<i>TotalBsmtSF</i>	$\hat{Y} = 279.70 + 0.124X$	2.2×10^{-16}	0.385	
<i>LotFrontage</i>	$\hat{Y} = 284.99 + 1.773X$	2.2×10^{-16}	0.132	

$\log(\text{LotArea})$	$\hat{Y} = -536.97 + 103.687X$	2.2×10^{-16}	0.153	
TotalFloorSF	$\hat{Y} = 233.00 + 0.120X$	2.2×10^{-16}	0.476	

The p-value of all 5 variables (2.2×10^{-16}) indicates a statistically significant relationship between each of them and $\sqrt{\text{SalePrice}}$ at 0.05 level of significance. Comparing the R^2 values, TotalFloorSF has the highest R^2 value of 0.476, indicating that it explains 47.6% of the variation in $\sqrt{\text{SalePrice}}$. However, looking at R^2 alone does not translate to a good model, especially since linear regression assumes normally distributed residuals, and according to the qq-plots above, TotalFloorSF 's model has residuals that deviate the most from normality. Therefore, taking into consideration both R^2 and the residual plot, TotalBsmfSF would be the best linear regression model for $\sqrt{\text{SalePrice}}$.

4.3 Principal Component Analysis (PCA)

In this section, we conduct Principal Component Analysis on our data. The crux of PCA lies in transforming the data into principal components (PCs) consisting of weighted existing continuous variables that account for the most variance. Doing so helps us to identify potential clusters within the data by uncovering where the data varies the most.

Importance of components:						
	PC1	PC2	PC3	PC4	PC5	PC6
Standard deviation	1.838	1.0201	0.9444	0.66336	0.49433	0.07290
Proportion of Variance	0.563	0.1734	0.1486	0.07334	0.04073	0.00089
Cumulative Proportion	0.563	0.7364	0.8850	0.95839	0.99911	1.00000

As we can see, PCA transforms our data into components PC1 to PC6, with PC1 to PC3 alone able to account for 88.5% of the total variance. We will therefore primarily investigate the contributors of PC1, 2 and 3.

	PC1	PC2	PC3
<i>sqrt(SalePrice)</i>	-0.4648636	-0.1876558	0.27837263
<i>TotalBsmtSF</i>	-0.3189359	-0.0570199	0.81900446
<i>log(LotArea)</i>	-0.3409947	0.5932416	-0.12898259
<i>GrLivArea</i>	-0.4773412	-0.3210890	-0.34394961
<i>LotFrontage</i>	-0.3269365	0.6375667	-0.06449653
<i>TotalFloorSF</i>	-0.4807933	-0.3162420	-0.33562289

A look at the rotation shows us the magnitude and direction of which the original continuous variables contribute to each PC. Contributing variables with the same sign move in the same direction, and larger magnitudes indicate a heavier weighting on the variable:

- **PC1 (56.3% of variance):** Its largest contributors are *sqrt(SalePrice)*, *GrLivArea* and *TotalFloorSF*. It captures a continuum from small and cheap homes to large and expensive homes.
- **PC2 (17.3% of variance):** Its largest contributors are *log(LotArea)* and *LotFrontage*, moving in the opposite direction to the other continuous variables. It captures a continuum from small homes built on large plots of land to large homes built on small plots of land.
- **PC3 (14.9% of variance):** Its largest contributors are *TotalBsmtSF* and *GrLivArea*, which move in opposite directions. It captures a continuum from homes with large above-ground living spaces but small basements to those with small above-ground living spaces but large basements.

Our findings of PC1's loadings enrich our insights from [Section 4.1](#) - not only do we know now that *sqrt(SalePrice)*, *GrLivArea* and *TotalFloorSF* have a stronger linear correlation, but we also know that this correlation between them accounts for 56.3% of the total variance in our data. Similarly, our findings of PC2 and PC3's loadings can be used to guide further studies of how and why different spatial configurations may influence sale prices.

5. Conclusion and Discussion

As we move into the modern age, housing has evolved significantly, shifting from its original purpose as a shelter, to become a centrepiece of integrated communities and societies. To that end, houses have been designed with various unique characteristics, such as different amenities, materials and sizes - which have in turn, caused them to be priced in increasingly complex ways.

By using an analysis technique called PCA on the prices of homes in Ames, Iowa, we have found the relationship of essential features like heating quality and control, alongside non-essential features like alley access and garage finishes to home prices. The findings that affected the overall "quality" and pricing of a house can be summarised in the following below:

- Quality of essential features such as heating, correlates with the quality of non-essential features such as garages
- Houses with paved alleys commanding a higher price than those with gravel alleys
- Type of roofs affects houses' sale price, with hip and mansard roofs being the most expensive due to weather conditions
- Year sold, on the other hand, has no relation to sale price

However, the accuracy of the analysis is limited, due to the dataset only being limited to Ames, Iowa. Further development of the analysis could be done by including additional housing data samples from other parts of the U.S. and integrating it with more advanced techniques like machine learning. This would allow us to explore the possibility of creating a pricing simulator, to support not just real estate developers, but also home buyers in finding a fair market price for a house in the near future.

6. Appendix

#3 Importing & Cleaning of Dataset

```
# Load dataset
house_data <- read.csv("train.csv")

# Convert YrSold to categorical
house_data$YrSold <- as.factor(house_data$YrSold)

# Check structure and summary
str(house_data)
summary(house_data)

# Discrete Numerical Variables :
# - OverallQual: Overall material and finish quality
# - GarageCars: Number of cars the garage can hold
# - Fireplaces: Number of fireplaces
# - KitchenAbvGr: Number of kitchens above ground
# - YrSold: Year the house was sold

# Continuous Numerical Variables :
# - SalePrice: Target variable (House price)
# - GrLivArea: Above ground living area (sqft)
# - TotalBsmntSF: Total basement area (sqft)
# - LotFrontage: Length of the street connected to the property
# - LotArea: Lot size in square feet
# - TotalFloorSF: Combined area of 1st and 2nd floor

# Categorical Variables :
# - HeatingQC: Heating quality and condition
# - RoofStyle: Type of roof
# - GarageFinish: Interior finish of the garage
# - BsmntFinType1: Rating of basement finished area
# - YearBuiltDecade: Decade the house was built
# - Alley: Type of alley

# Create working dataframe with selected variables (excluding X1stFlrSF)
df <- house_data[, c("OverallQual", "GarageCars", "Fireplaces", "LotFrontage", "YearBuilt",
"YrSold",
                    "SalePrice", "GrLivArea", "TotalBsmntSF", "KitchenAbvGr",
                    "HeatingQC", "RoofStyle", "GarageFinish", "BsmntFinType1", "Alley")]

# Add TotalFloorSF (1st Floor SF + 2nd Floor SF)
df$TotalFloorSF <- house_data$X1stFlrSF + house_data$X2ndFlrSF

# Add log(LotArea)
df$log(LotArea) <- log(house_data$LotArea)

#Add sqrt(SalePrice)
df$`sqrt(SalePrice)` <- sqrt(house_data$SalePrice)
```

```

# Create YearBuiltDecade as a new categorical variable
df$YearBuiltDecade <- cut(df$YearBuilt,
                          breaks = seq(1870, 2020, by = 10),
                          right = FALSE,
                          labels = paste(seq(1870, 2010, by = 10), "s", sep = ""))

# Define variable groups
all_vars <- c("OverallQual", "GarageCars", "Fireplaces", "LotFrontage", "YrSold", "SalePrice",
              "sqrt(SalePrice)", "GrLivArea", "TotalBsmtSF", "KitchenAbvGr", "log(LotArea)",
              "TotalFloorSF",
              "HeatingQC", "RoofStyle", "GarageFinish", "BsmtFinType1", "YearBuiltDecade",
              "Alley")

discrete_vars <- c("OverallQual", "GarageCars", "Fireplaces", "KitchenAbvGr", "YrSold")

continuous_vars <- c("sqrt(SalePrice)", "GrLivArea", "TotalBsmtSF", "LotFrontage",
                    "log(LotArea)", "TotalFloorSF")

cat_vars <- c("HeatingQC", "RoofStyle", "GarageFinish", "BsmtFinType1", "YearBuiltDecade",
              "Alley")

```

#3.1 Removing incomplete data from dataset

```

# Replace NA in GarageFinish where NA means "No Garage"
df$GarageFinish[is.na(df$GarageFinish)] <- "NoGarage"

# Replace NA in BsmtFinType1 where NA means "No Basement"
df$BsmtFinType1[is.na(df$BsmtFinType1)] <- "NoBasement"

# Replace NA in Alley where NA means "No Alley"
df$Alley[is.na(df$Alley)] <- "NoAlley"

# Check for missing values
colSums(is.na(df))

# Remove rows with NA
df <- na.omit(df)
str(df)

# Outlier removal function
remove_outliers <- function(x) {
  q1 <- quantile(x, 0.25, na.rm = TRUE)
  q3 <- quantile(x, 0.75, na.rm = TRUE)
  iqr <- q3 - q1
  lower_bound <- q1 - 1.5 * iqr
  upper_bound <- q3 + 1.5 * iqr
  x[x < lower_bound | x > upper_bound] <- NA
  return(x)
}

```

#3.2 Summary statistics for the main variable of interest, SalePrice

```

#Histogram of SalePrice
hist(df$SalePrice, main = "Histogram of SalePrice", xlab = "SalePrice")

```

```
#Histogram of sqrt(SalePrice)
hist(df$`sqrt(SalePrice)` , main ="Histogram of sqrt(SalePrice)", xlab = "sqrt(SalePrice)")
```

```
#Boxplot of sqrt(SalePrice)
boxplot(df$`sqrt(SalePrice)` , main ="Histogram of sqrt(SalePrice)")
```

```
#Histogram of SalePrice
hist(df$SalePrice, main ="Histogram of SalePrice", xlab = "SalePrice")
```

#3.3 Summary statistics for continuous variables

```
#Histogram of GrLivArea
hist(df$GrLivArea, main ="Histogram of GrLivArea", xlab = "GrLivArea")
```

```
#Histogram of TotalBsmtSF
hist(df$TotalBsmtSF, main ="Histogram of TotalBsmtSF", xlab = "TotalBsmtSF")
```

```
#Histogram of LotFrontage
hist(df$LotFrontage, main ="Histogram of LotFrontage", xlab = "LotFrontage")
```

```
#Histogram of log(LotArea)
hist(df$log(LotArea)` , main ="Histogram of log(LotArea)", xlab = "log(LotArea)")
```

```
#Histogram of TotalFloorSF
hist(df$TotalFloorSF, main ="Histogram of TotalFloorSF", xlab = "TotalFloorSF")
```

```
#Boxplot of SalePrice
boxplot(df$SalePrice,main ="Boxplot of SalePrice")
```

```
#Boxplot of GrLivArea
boxplot(df$GrLivArea,main ="Boxplot of GrLivArea")
```

```
#Boxplot of TotalBsmtSF
boxplot(df$TotalBsmtSF,main ="Boxplot of TotalBsmtSF")
```

```
#Boxplot of LotFrontage
boxplot(df$LotFrontage,main ="Boxplot of LotFrontage")
```

```
#Boxplot of LotArea
boxplot(df$log(LotArea)` ,main ="Boxplot of log(LotArea)")
```

```
#Boxplot of TotalFloorSF
boxplot(df$TotalFloorSF,main ="Boxplot of TotalFloorSF")
```

#3.4 Summary statistics for discrete variables

```
#Barplot of OverallQual
barplot(table(df$OverallQual),
      main = "Barplot of OverallQual",
      xlab = "Overall Quality",
      ylab = "Frequency")
```

```
#Barplot of GarageCars
```

```

barplot(table(df$GarageCars),
        main = "Barplot of GarageCars",
        xlab = "Garage Cars",
        ylab = "Frequency")

#Barplot of Fireplaces
barplot(table(df$Fireplaces),
        main = "Barplot of Fireplaces",
        xlab = "Fireplaces",
        ylab = "Frequency")

#Barplot of KitchenAbvGr
barplot(table(df$KitchenAbvGr),
        main = "Barplot of KitchenAbvGr",
        xlab = "Number of Kitchen Above Grade",
        ylab = "Frequency")

#Barplot of YrSold
barplot(table(df$YrSold),
        main = "Barplot of YrSold",
        xlab = "Year Sold",
        ylab = "Frequency")

```

#3.5 Summary statistics for the categorical variables

```

#Barplot of HeatingQC
barplot(table(df$HeatingQC),
        main = "Barplot of HeatingQC",
        xlab = "HeatingQC",
        ylab = "Frequency")

#Barplot of RoofStyle
barplot(table(df$RoofStyle),
        main = "Barplot of RoofStyle",
        xlab = "RoofStyle",
        ylab = "Frequency")

#Barplot of GarageFinish
barplot(table(df$GarageFinish),
        main = "Barplot of GarageFinish",
        xlab = "GarageFinish",
        ylab = "Frequency")

#Barplot of BsmtFinType1
barplot(table(df$BsmtFinType1),
        main = "Barplot of BsmtFinType1",
        xlab = "BsmtFinType",
        ylab = "Frequency")

#Barplot of YearBuiltDecade
barplot(table(df$YearBuiltDecade),
        main = "Barplot of YearBuiltDecade",
        xlab = "YearBuiltDecade",
        ylab = "Frequency")

```

```
#Barplot of Alley (without NoAlley)
barplot(table(df$Alley[df$Alley != "NoAlley"]),
        main = "Barplot of Alley",
        xlab = "Alley",
        ylab = "Frequency")
```

```
# Apply outlier removal to continuous variables
df_outlier_filtered <- df
for (col in continuous_vars) {
  df_outlier_filtered[[col]] <- remove_outliers(df_outlier_filtered[[col]])
}
```

#3.6 Summary statistics for all variables

```
# Check for number of outliers in respective columns
colSums(is.na(df_outlier_filtered))
```

```
# Remove rows with any NA (from outliers)
filtered_df <- df_outlier_filtered[complete.cases(df_outlier_filtered), ]
```

```
# Revised Histogram of sqrt(SalePrice)
hist(filtered_df$`sqrt(SalePrice)` , main ="Histogram of sqrt(SalePrice) (outliers removed)",
      xlab = "sqrt(SalePrice)")
```

```
# Revised Boxplot of sqrt(SalePrice)
boxplot(df$`sqrt(SalePrice)` ,main ="Boxplot of sqrt(SalePrice) (outliers removed)",outline=FALSE)
```

```
# Summary for sqrt(SalePrice) after removing outliers
summary(filtered_df$`sqrt(SalePrice)`)
```

```
# Check structure and summary
str(filtered_df)
summary(filtered_df)
```

#4.1 Correlations between sqrt(SalePrice) and other Continuous Variables

```
# Load necessary libraries
library(ggplot2)
library(ggcorrplot)
```

```
# Select relevant columns
df_subset <- filtered_df[, c("sqrt(SalePrice)", "TotalBsmtSF", "log(LotArea)", "GrLivArea",
"LotFrontage", "TotalFloorSF")]
```

```
# Compute correlation matrix
cor_matrix <- cor(df_subset, use = "complete.obs")
```

```
# Plot correlation heatmap
ggcorrplot(cor_matrix,
           method = "circle",
           type = "lower",
```

```
lab = TRUE,
lab_size = 3,
colors = c("blue", "white", "red"),
title = "Correlation Heatmap with sqrt(SalePrice)",
ggtheme = theme_minimal())
```

#4.2.1 Relation between sqrt(SalePrice) and OverallQual

```
#Plotting boxplot
boxplot(filtered_df$`sqrt(SalePrice)`~filtered_df$OverallQual)
```

#4.2.2 Relation between HeatingQC and GarageFinish

```
#Plotting two-way contingency table (GarageFinish and HeatingQC)
BsmtFinType1vsHeatingQC <- table(filtered_df$GarageFinish, filtered_df$HeatingQC)
```

```
#Checking if table is generated correctly
print(BsmtFinType1vsHeatingQC)
```

```
#Removing the 'Po' column
BsmtFinType1vsHeatingQC_filtered <- BsmtFinType1vsHeatingQC[,
colnames(BsmtFinType1vsHeatingQC) != "Po"]
```

```
#Checking if table is generated correctly
print(BsmtFinType1vsHeatingQC_filtered)
```

```
#Conducting chi-square test of independence
chisq.test(BsmtFinType1vsHeatingQC_filtered)
```

#Since the p-value<0.05, we conclude that there likely exists an association between the quality of the basement finished area and the quality of the garage finish

#4.2.3 Relation between sqrt(SalePrice) and Alley

```
df_alley <- filtered_df[filtered_df$Alley != "NoAlley", ]
```

```
#Variances are completely unknown to us so we compare using f-test! (var.test)
var.test(`sqrt(SalePrice)` ~ Alley, data = df_alley, conf.level = 0.9)
```

```
#Reject h0, take it as variances not equal
t.test(`sqrt(SalePrice)` ~ Alley, data = df_alley, var.equal=FALSE)
```

#4.2.4 Relation between sqrt(SalePrice) and RoofStyle

```
#Boxplot of sqrt(SalePrice) by RoofStyle, in order of increasing sqrt(SalePrice)
reordered_roofstyle <- reorder(filtered_df$RoofStyle, filtered_df$`sqrt(SalePrice)` , FUN =
median)
boxplot(filtered_df$`sqrt(SalePrice)` ~ reordered_roofstyle, main = "sqrt(SalePrice) by
RoofStyle", xlab = "RoofStyle", ylab = "sqrt(SalePrice)", col = "skyblue", las = 2)
```

```
#ANOVA for RoofStyle and sqrt(SalePrice)
roofstyle_anova <- aov(`sqrt(SalePrice)` ~ RoofStyle, data = filtered_df)
summary(roofstyle_anova)
#pvalue < 0.05, so sale price is related to roof style
```

```

#Display number of houses in each RoofStyle
table(filtered_df$RoofStyle)
#Kruskal-Willies Test
kruskal.test(`sqrt(SalePrice)` ~ RoofStyle, data = filtered_df)
#pvalue < 0.05, suggesting that sale price is related to roof style

#4.2.5 Relation between sqrt(SalePrice) and YrSold
#Plotting boxplot
boxplot(filtered_df$`sqrt(SalePrice)`~filtered_df$YrSold)

#Plotting one-way ANOVA model
summary(aov(filtered_df$`sqrt(SalePrice)`~filtered_df$YrSold))

##pvalue>0.05 so saleprice is independent of year sold
#BUT, looking at boxplot, explain why this may be otherwise as well

#4.2.6 Linear regression of continuous variables against sqrt(SalePrice)

# Create linear regression models
model1<-lm(filtered_df$`sqrt(SalePrice)` ~ filtered_df$GrLivArea)
model2<-lm(filtered_df$`sqrt(SalePrice)` ~ filtered_df$TotalBsmtSF)
model3<-lm(filtered_df$`sqrt(SalePrice)` ~ filtered_df$LotFrontage)
model4<-lm(filtered_df$`sqrt(SalePrice)` ~ filtered_df$log(LotArea))
model5<-lm(filtered_df$`sqrt(SalePrice)` ~ filtered_df$TotalFloorSF)

# Summary of models
summary(model1)
summary(model2)
summary(model3)
summary(model4)
summary(model5)

par(mfrow = c(2, 3))

# Plot sqrt(SalePrice) against GrLivArea
plot(filtered_df$GrLivArea, filtered_df$`sqrt(SalePrice)` , xlab = "GrLivArea", ylab =
"sqrt(SalePrice)")
abline(model1, col = "red")

# Plot sqrt(SalePrice) against TotalBsmtSF
plot(filtered_df$TotalBsmtSF, filtered_df$`sqrt(SalePrice)` , xlab = "TotalBsmtSF", ylab =
"sqrt(SalePrice)")
abline(model2, col = "blue")

# Plot sqrt(SalePrice) against LotFrontage
plot(filtered_df$LotFrontage, filtered_df$`sqrt(SalePrice)` , xlab = "LotFrontage", ylab =
"sqrt(SalePrice)")
abline(model3, col = "green")

# Plot sqrt(SalePrice) against log(LotArea)
plot(filtered_df$log(LotArea), filtered_df$`sqrt(SalePrice)` , xlab = "log(LotArea)", ylab =
"sqrt(SalePrice)")
abline(model4, col = "purple")

```

```

# Plot sqrt(SalePrice) against TotalFloorSF
plot(filtered_df$TotalFloorSF, filtered_df$`sqrt(SalePrice)` , xlab = "TotalFloorSF", ylab =
"sqrt(SalePrice)")
abline(model5, col = "orange")

par(mfrow = c(2, 3))

# qqplot of residuals
qqnorm(model1$residuals, main = "Model 1")
qqline(model1$residuals, col = "blue")
qqnorm(model2$residuals, main = "Model 2")
qqline(model2$residuals, col = "blue")
qqnorm(model3$residuals, main = "Model 3")
qqline(model3$residuals, col = "blue")
qqnorm(model4$residuals, main = "Model 4")
qqline(model4$residuals, col = "blue")
qqnorm(model5$residuals, main = "Model 5")
qqline(model5$residuals, col = "blue")

#4.3 Principal Component Analysis (PCA)
# Select relevant columns
df_subset <- filtered_df[, c("sqrt(SalePrice)", "TotalBsmtSF", "log(LotArea)", "GrLivArea",
"LotFrontage", "TotalFloorSF")]

# PCA Analysis
num_scaled <- scale(df_subset)
pca_result <- prcomp(num_scaled, center = TRUE, scale. = TRUE)

# Summary of variance explained
summary(pca_result)

# Checking variable loadings
pca_result$rotation[, 1:3]

```

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